

```

In [6]: import numpy as np
import matplotlib.pyplot as plt

t = np.linspace(0, 2*np.pi, 100)
omega = 2
x1 = 0.5 * np.sin(omega * t)
x2 = np.cos(omega * t)

# Plot 1: Position and Velocity vs Time
plt.figure(1)
plt.plot(t, x1, label='Position (x1)')
plt.plot(t, x2, label='Velocity (x2)')
plt.xlabel('Time (t)')
plt.ylabel('Value')
plt.title('Position and Velocity vs Time')
plt.legend()
plt.grid()

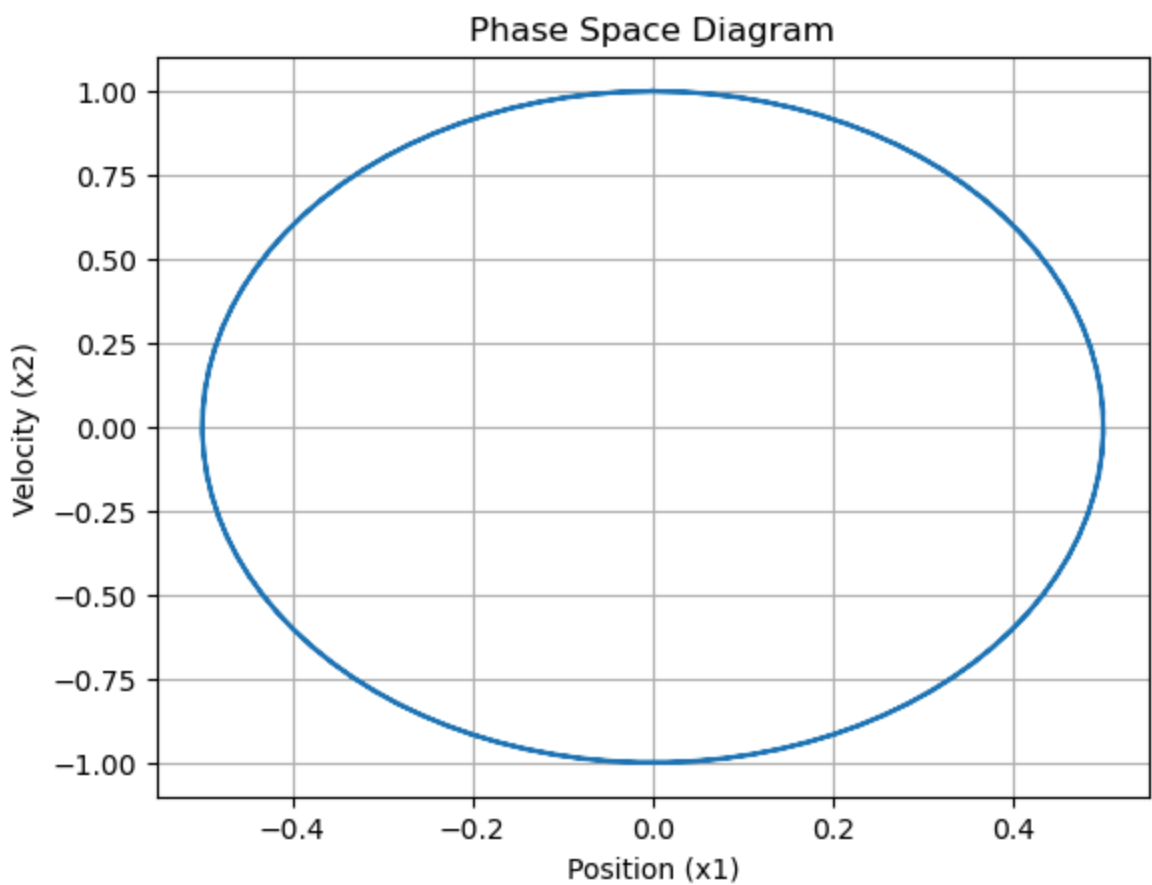
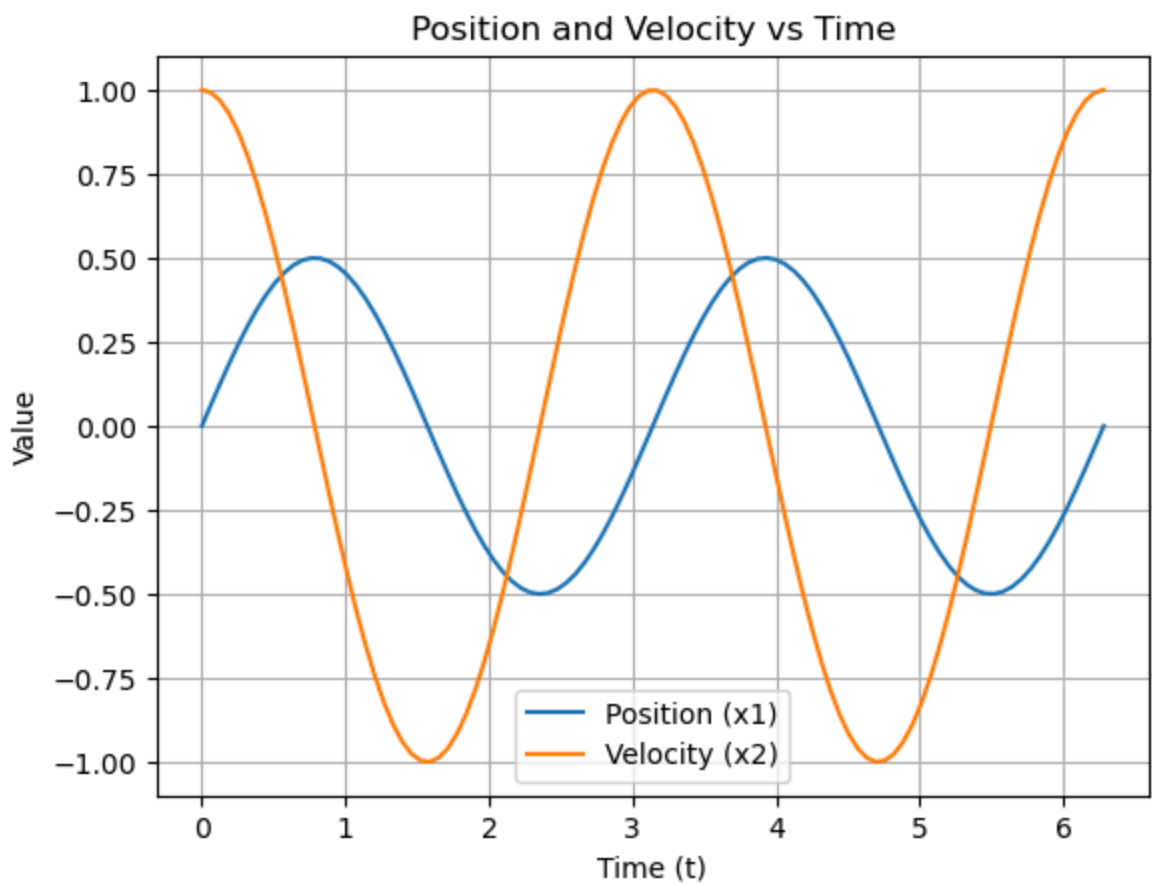
# Plot 2: Phase Space Diagram
plt.figure(2)
plt.plot(x1, x2)
plt.xlabel('Position (x1)')
plt.ylabel('Velocity (x2)')
plt.title('Phase Space Diagram')
plt.grid()

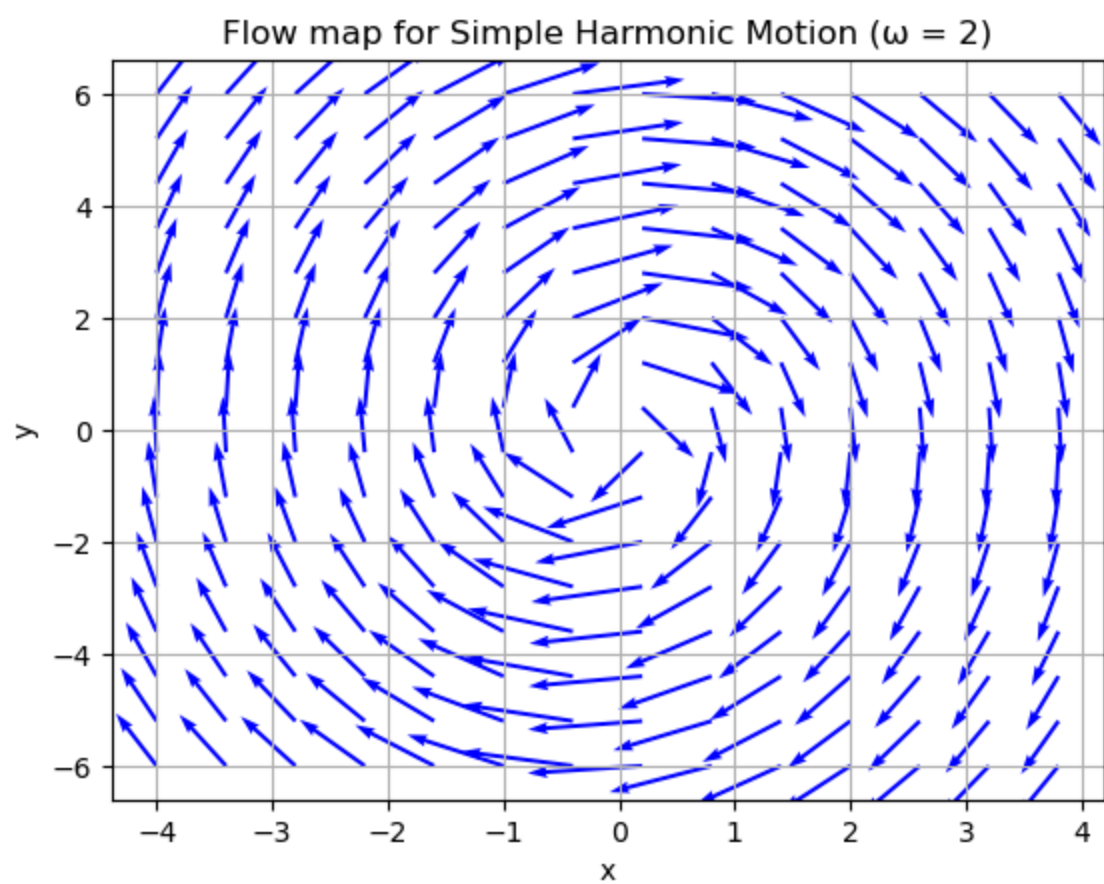
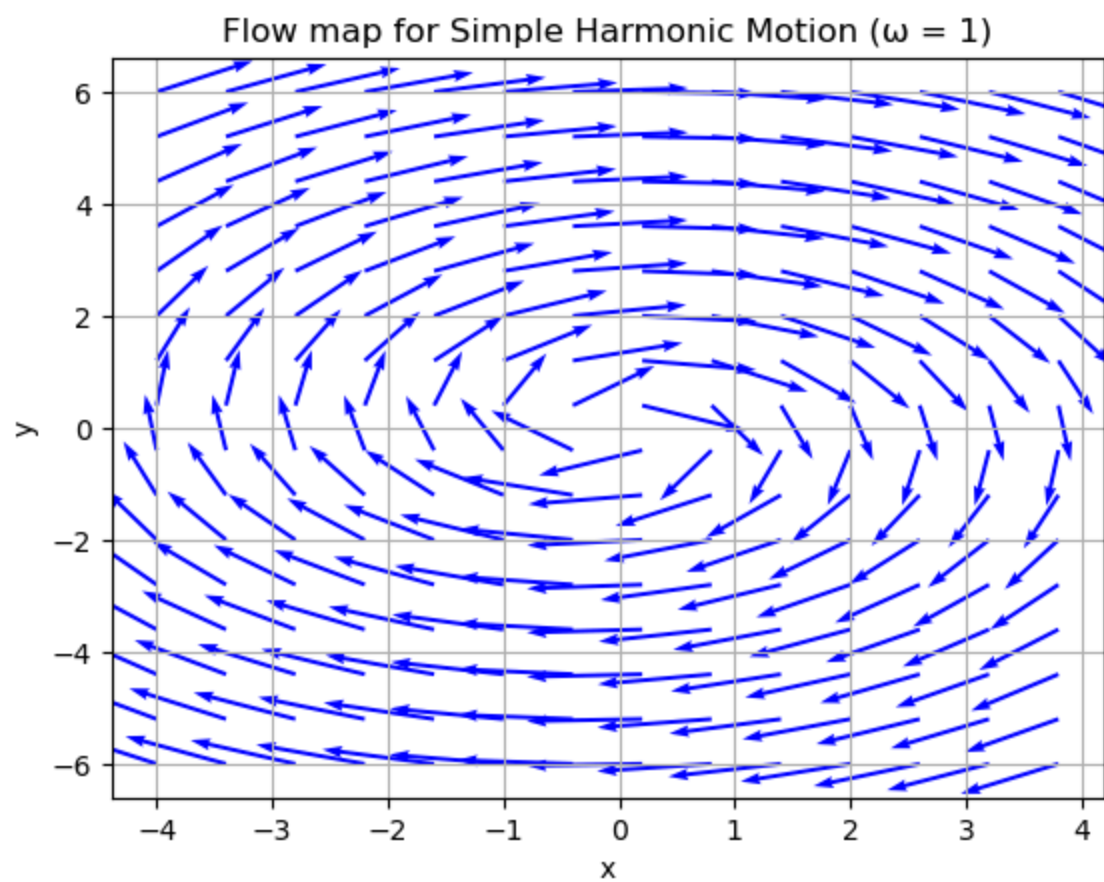
# Plot 3: Flow Map for different omega values
omegas = [1, 2, 3]
x = np.arange(-4, 4.1, 0.6)
y = np.arange(-6, 6.1, 0.8)
xv, yv = np.meshgrid(x, y)

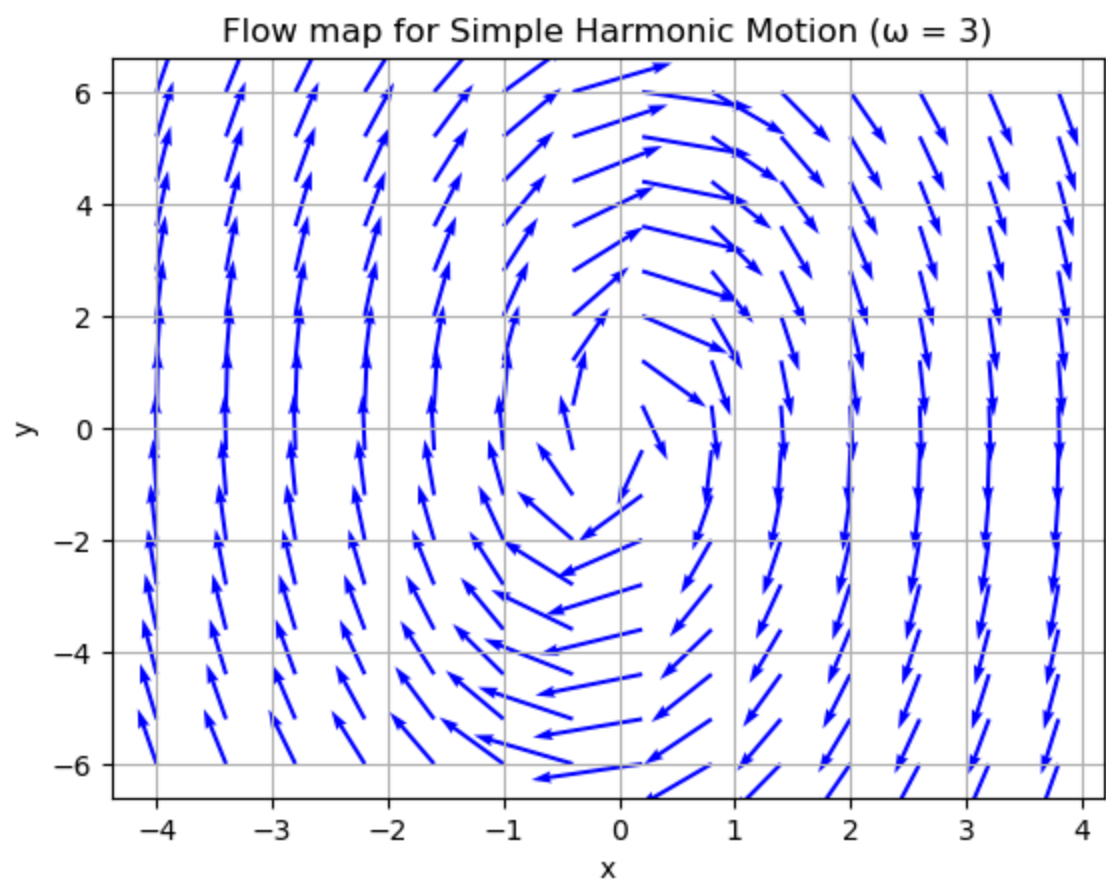
for w in omegas:
    dx = yv
    dy = -w**2 * xv
    mag = np.sqrt(dx**2 + dy**2)
    dxu = dx / mag
    dyu = dy / mag

    plt.figure(3)
    plt.quiver(xv, yv, dxu, dyu, scale=1, scale_units='xy', angles='xy', width=0.005)
    plt.xlabel('x')
    plt.ylabel('y')
    plt.title(f'Flow map for Simple Harmonic Motion ( $\omega = \{w\}$ )')
    plt.grid()
    plt.show()

```







In []: